

Exploratory Data Analysis

1. impute and compute state average. make sure account for survey design by incorporating the weights
2. create a time series database of state policies after June 24, 2022 where the categorical variable has n categories

```
library(haven)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.2      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
dat25 <- read_xpt("../data/LLCP2025.XPT")
dat24 <- read_xpt("../data/LLCP2024.XPT")
#dat23 <- read_xpt("../data/LLCP2023.XPT")
dat22 <- read_xpt("../data/LLCP2022.XPT")
#dat21 <- read_xpt("../data/LLCP2021.XPT")
#dat20 <- read_xpt("../data/LLCP2020.XPT")
dat19 <- read_xpt("../data/LLCP2019.XPT")
#dat18 <- read_xpt("../data/LLCP2018.XPT")
dat17 <- read_xpt("../data/LLCP2017.XPT")
```

Exploratory data analysis

variables asked only in 2025, 2024, 2022, 2019, and 2017

NOBCUSE9 = What was main reason for not doing anything to keep you from getting pregnant?

PFPPRVN4 = The last time you had sexual intercourse, did you or your partner do anything to keep you from getting pregnant? () FAMILY PLANNING = The last time you had sexual intercourse, what did you or your partner do to keep you from getting pregnant? (lots of response categories!!!)

variables asked in all surveys

SEXVAR = 2 if female

_AGEG5YR = 5-year age groups

EDUCA = What is the highest grade or year of school you completed? (1=no high school, 2=high school, 3=some college, 4=college, 9=don't know)

MARITAL= marital status (1=married, 2=divorced, 3=widowed, 4=separated, 5=never married, 6=A member of an unmarried couple, 9 =refused)

HOME= homeowner status (1=own, 2=rent, 3=other arrangement, 7=don't know, 9=refused)

PREGNANT = current pregnancy status (1=yes, 2=no, 7=don't know, 9=refused)

MENTHLTH = Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good? (1-30, 88=none, 77=don't know, 99=refused) - consider state-level databases for COVID-19

EMPLOY = employment status (1 = employed for wages, 2 = self-employed, 3 = out of work for 1 year or more, 4 = out of work for less than 1 year, 5 = homemaker, 6 = student, 7 = retired, 8 = unable to work, 9 = refused)

INCOME = income categories (1=less than 15k, 2=15k to 25k, 3= 25k to 35k, 4= 35k to 50k, 5=50k to 100k, 6=100k to 200k, 7=200k or more, 9=don't know)

INSURANCE = Respondents aged 18-64 who have any form of health insurance (1=yes, 2=no, 9=don't know) - consider finding a database that focuses on Medicaid access at the state level and use this as a confounder to address omitted variable bias at the state level

RACE = Race/ethnicity categories (1=nh white, 2=nh black, 3=nh native american, 4=nh asian, 5=nh native hawaiian, 7=nh multiracial, 9=don't know)

subset each survey year to include relevant variables, subsetting the sample to only women

Recode the states in each year since states that didn't collect enough data for a given year weren't included in the dataset

```
sub_dat25 <- dat25 |> select(c("_STATE", "_AGEG5YR", "SEXVAR", "PREGNANT",  
  ↪ "MENTHLTH", "EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU654", "_RACE",  
  ↪ "RENTHOM1", "MARITAL", "TYPCNT10")) |> filter(SEXVAR==2) |>  
  ↪ select(-SEXVAR) |> rename(INCOME='_INCOMG1', INSURANCE='_HCVU654',  
  ↪ EMPLOY='EMPLOY1', FAMILY_PLANNING="TYPCNT10", "HOME"=RENTHOM1,  
  ↪ RACE="_RACE")  
  
state_labels_2025 <- c(  
  `1`  = "Alabama",  
  `2`  = "Alaska",  
  `4`  = "Arizona",  
  `5`  = "Arkansas",  
  `8`  = "Colorado",  
  `9`  = "Connecticut",  
  `10` = "Delaware",  
  `11` = "District of Columbia",  
  `12` = "Florida",  
  `13` = "Georgia",  
  `15` = "Hawaii",  
  `16` = "Idaho",  
  `17` = "Illinois",  
  `18` = "Indiana",  
  `19` = "Iowa",  
  `20` = "Kansas",  
  `21` = "Kentucky",  
  `22` = "Louisiana",  
  `23` = "Maine",  
  `24` = "Maryland",  
  `25` = "Massachusetts",  
  `26` = "Michigan",  
  `27` = "Minnesota",  
  `29` = "Missouri",  
  `30` = "Montana",  
  `31` = "Nebraska",  
  `33` = "New Hampshire",  
  `34` = "New Jersey",  
  `35` = "New Mexico",
```

```

`36` = "New York",
`37` = "North Carolina",
`38` = "North Dakota",
`39` = "Ohio",
`40` = "Oklahoma",
`41` = "Oregon",
`42` = "Pennsylvania",
`44` = "Rhode Island",
`45` = "South Carolina",
`46` = "South Dakota",
`47` = "Tennessee",
`48` = "Texas",
`49` = "Utah",
`50` = "Vermont",
`51` = "Virginia",
`53` = "Washington",
`54` = "West Virginia",
`55` = "Wisconsin",
`56` = "Wyoming",
`66` = "Guam",
`72` = "Puerto Rico"
)

sub_dat25 <- sub_dat25 |>
  dplyr::mutate(
    `_STATE` = as.character(unname(
      state_labels_2025[as.character(`_STATE`)]
    ))
  ) |>
  mutate(EDUCA = case_when(
    EDUCA %in% 1:3 ~ 1, # no high school
    EDUCA == 4 ~ 2, # high school
    EDUCA == 5 ~ 3, #Some college
    EDUCA == 6 ~ 4, #College graduate
    EDUCA == 9 ~ 9 #Don't know
  ))

```

```

sub_dat24 <- dat24 |> select(c("_STATE", "_AGEG5YR", "SEXVAR", "PREGNANT",
  ↪ "MENTHLTH", "EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU654", "_RACE",
  ↪ "RENTHOM1", "MARITAL", "TYPCNTR9")) |> filter(SEXVAR==2) |>
  ↪ select(-SEXVAR) |> rename(INCOME='_INCOMG1', INSURANCE='_HCVU654',
  ↪ EMPLOY='EMPLOY1', FAMILY_PLANNING="TYPCNTR9", "HOME"=RENTHOM1,
  ↪ RACE="_RACE")

```

```
state_labels_2024 <- c(
  `1` = "Alabama",
  `2` = "Alaska",
  `4` = "Arizona",
  `5` = "Arkansas",
  `6` = "California",
  `8` = "Colorado",
  `9` = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
  `21` = "Kentucky",
  `22` = "Louisiana",
  `23` = "Maine",
  `24` = "Maryland",
  `25` = "Massachusetts",
  `26` = "Michigan",
  `27` = "Minnesota",
  `28` = "Mississippi",
  `29` = "Missouri",
  `30` = "Montana",
  `31` = "Nebraska",
  `32` = "Nevada",
  `33` = "New Hampshire",
  `34` = "New Jersey",
  `35` = "New Mexico",
  `36` = "New York",
  `37` = "North Carolina",
  `38` = "North Dakota",
  `39` = "Ohio",
  `40` = "Oklahoma",
  `41` = "Oregon",
  `42` = "Pennsylvania",
  `44` = "Rhode Island",
  `45` = "South Carolina",
```

```

`46` = "South Dakota",
`48` = "Texas",
`49` = "Utah",
`50` = "Vermont",
`51` = "Virginia",
`53` = "Washington",
`54` = "West Virginia",
`55` = "Wisconsin",
`56` = "Wyoming",
`66` = "Guam",
`72` = "Puerto Rico",
`78` = "Virgin Islands"
)
sub_dat24 <- sub_dat24 |>
  dplyr::mutate(
    `_STATE` = as.character(unnamed(
      state_labels_2024[as.character(`_STATE`)]
    ))
  ) |> mutate(EDUCA = case_when(
    EDUCA %in% 1:3 ~ 1, # no high school
    EDUCA == 4 ~ 2, # high school
    EDUCA == 5 ~ 3, #Some college
    EDUCA == 6 ~ 4, #College graduate
    EDUCA == 9 ~ 9 #Don't know
  ))

```

```

# Roe v. Wade was overturned on June 24, 2022
sub_dat22 <- dat22 |>
  ↪ select(c("_STATE", "_AGE5YR", "SEXVAR", "IMONTH", "IDAY", "IYEAR", "PREGNANT",
  ↪ "MENTHLTH", "EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU652", "_RACE1",
  ↪ "TYPCNTR9", "RENTHOM1")) |> filter(SEXVAR==2) |> select(-SEXVAR) |>
  ↪ rename(INCOME='_INCOMG1', INSURANCE='_HCVU652', EMPLOY='EMPLOY1',
  ↪ FAMILY_PLANNING="TYPCNTR9", HOME="RENTHOM1", RACE="_RACE1")
state_labels_2022 <- c(
  `1` = "Alabama",
  `2` = "Alaska",
  `4` = "Arizona",
  `5` = "Arkansas",
  `6` = "California",
  `8` = "Colorado",
  `9` = "Connecticut",
  `10` = "Delaware",

```

```
`11` = "District of Columbia",
`12` = "Florida",
`13` = "Georgia",
`15` = "Hawaii",
`16` = "Idaho",
`17` = "Illinois",
`18` = "Indiana",
`19` = "Iowa",
`20` = "Kansas",
`21` = "Kentucky",
`22` = "Louisiana",
`23` = "Maine",
`24` = "Maryland",
`25` = "Massachusetts",
`26` = "Michigan",
`27` = "Minnesota",
`28` = "Mississippi",
`29` = "Missouri",
`30` = "Montana",
`31` = "Nebraska",
`32` = "Nevada",
`33` = "New Hampshire",
`34` = "New Jersey",
`35` = "New Mexico",
`36` = "New York",
`37` = "North Carolina",
`38` = "North Dakota",
`39` = "Ohio",
`40` = "Oklahoma",
`41` = "Oregon",
`42` = "Pennsylvania",
`44` = "Rhode Island",
`45` = "South Carolina",
`46` = "South Dakota",
`47` = "Tennessee",
`48` = "Texas",
`49` = "Utah",
`50` = "Vermont",
`51` = "Virginia",
`53` = "Washington",
`54` = "West Virginia",
`55` = "Wisconsin",
`56` = "Wyoming",
```

```

`66` = "Guam",
`72` = "Puerto Rico",
`78` = "Virgin Islands"
)
sub_dat22 <- sub_dat22 |>
  dplyr::mutate(
    `_STATE` = as.character(unname(
      state_labels_2022[as.character(`_STATE`)]
    ))
  )|> mutate(EDUCA = case_when(
    EDUCA %in% 1:3 ~ 1, # no high school
    EDUCA == 4 ~ 2, # high school
    EDUCA == 5 ~ 3, #Some college
    EDUCA == 6 ~ 4, #College graduate
    EDUCA == 9 ~ 9 #Don't know
  ))

```

```

sub_dat19 <- dat19 |> select(c("_STATE", "_AGEG5YR", "SEXVAR", "PREGNANT", "MENT",
  ↪ HLTH", "EMPLOY1", "_INCOMG", "EDUCA", "_HCVU651", "_RACE", "RENTHOM1",
  ↪ "MARITAL", "TYPCNTR8")) |> filter(SEXVAR==2) |> select(-SEXVAR) |>
  ↪ rename(INCOME='_INCOMG', INSURANCE='_HCVU651', EMPLOY='EMPLOY1',
  ↪ FAMILY_PLANNING="TYPCNTR8", "HOME"=RENTHOM1, RACE="_RACE")
state_labels_2019 <- c(
  `1` = "Alabama",
  `2` = "Alaska",
  `4` = "Arizona",
  `5` = "Arkansas",
  `6` = "California",
  `8` = "Colorado",
  `9` = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
  `21` = "Kentucky",
  `22` = "Louisiana",

```

```

`23` = "Maine",
`24` = "Maryland",
`25` = "Massachusetts",
`26` = "Michigan",
`27` = "Minnesota",
`28` = "Mississippi",
`29` = "Missouri",
`30` = "Montana",
`31` = "Nebraska",
`32` = "Nevada",
`33` = "New Hampshire",
`35` = "New Mexico",
`36` = "New York",
`37` = "North Carolina",
`38` = "North Dakota",
`39` = "Ohio",
`40` = "Oklahoma",
`41` = "Oregon",
`42` = "Pennsylvania",
`44` = "Rhode Island",
`45` = "South Carolina",
`46` = "South Dakota",
`47` = "Tennessee",
`48` = "Texas",
`49` = "Utah",
`50` = "Vermont",
`51` = "Virginia",
`53` = "Washington",
`54` = "West Virginia",
`55` = "Wisconsin",
`56` = "Wyoming",
`66` = "Guam",
`72` = "Puerto Rico"
)
sub_dat19 <- sub_dat19 |>
  dplyr::mutate(
    `_STATE` = as.character(unname(
      state_labels_2019[as.character(`_STATE`)]
    ))
  )
sub_dat19 |>
  mutate(EDUCA = case_when(
    EDUCA %in% 1:3 ~ 1, # no high school
    EDUCA == 4 ~ 2, # high school
  ))

```

```

    EDUCA == 5 ~ 3 ,#Some college
    EDUCA == 6 ~ 4 ,#College graduate
    EDUCA == 9 ~ 9 #Don't know
  ))

```

```

sub_dat17 <- dat17 |> select(c("_STATE", "_AGEG5YR", "SEX","PREGNANT",
  ↪ "MENTHLTH", "EMPLOY1", "_INCOMG", "EDUCA", "_HCVU651", "_RACE",
  ↪ "RENTHOM1", "MARITAL", "TYPCNTR7")) |> filter(SEX==2) |> select(-SEX) |>
  ↪ rename(INCOME='_INCOMG', INSURANCE='_HCVU651', EMPLOY='EMPLOY1',
  ↪ FAMILY_PLANNING="TYPCNTR7", "HOME"=RENTHOM1, RACE="_RACE")
state_labels_2017 <- c(
  `1` = "Alabama",
  `2` = "Alaska",
  `4` = "Arizona",
  `5` = "Arkansas",
  `6` = "California",
  `8` = "Colorado",
  `9` = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
  `21` = "Kentucky",
  `22` = "Louisiana",
  `23` = "Maine",
  `24` = "Maryland",
  `25` = "Massachusetts",
  `26` = "Michigan",
  `27` = "Minnesota",
  `28` = "Mississippi",
  `29` = "Missouri",
  `30` = "Montana",
  `31` = "Nebraska",
  `32` = "Nevada",
  `33` = "New Hampshire",
  `34` = "New Jersey",

```

```

`35` = "New Mexico",
`36` = "New York",
`37` = "North Carolina",
`38` = "North Dakota",
`39` = "Ohio",
`40` = "Oklahoma",
`41` = "Oregon",
`42` = "Pennsylvania",
`44` = "Rhode Island",
`45` = "South Carolina",
`46` = "South Dakota",
`47` = "Tennessee",
`48` = "Texas",
`49` = "Utah",
`50` = "Vermont",
`51` = "Virginia",
`53` = "Washington",
`54` = "West Virginia",
`55` = "Wisconsin",
`56` = "Wyoming",
`66` = "Guam",
`72` = "Puerto Rico"
)
sub_dat17 <- sub_dat17 |>
  dplyr::mutate(
    `STATE` = as.character(unname(
      state_labels_2017[as.character(`STATE`)]
    ))
  )
mutate(EDUCA = case_when(
  EDUCA %in% 1:3 ~ 1, # no high school
  EDUCA == 4 ~ 2, # high school
  EDUCA == 5 ~ 3, #Some college
  EDUCA == 6 ~ 4, #College graduate
  EDUCA == 9 ~ 9 #Don't know
))

```

```

sub_dat23 <- dat23 |> select(c("_STATE", "SEXVAR", "PREGNANT",
  ↪ "MENTHLTH", "EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU653", "_RACE")) |>
  ↪ filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
  ↪ INSURANCE='_HCVU653', EMPLOY='EMPLOY1')

```

```

sub_dat21 <- dat21 |> select(c("_STATE","SEXVAR","PREGNANT",
  ↪ "MENTHLTH","EMPLOY1","_INCOMG1", "EDUCA","_HCVU652", "_RACE")) |>
  ↪ filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
  ↪ INSURANCE='_HCVU652', EMPLOY='EMPLOY1')

sub_dat20 <- dat20 |> select(c("_STATE","SEXVAR","PREGNANT",
  ↪ "MENTHLTH","EMPLOY1", "_INCOMG", "EDUCA", "_HCVU651", "_RACE")) |>
  ↪ filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG',
  ↪ INSURANCE='_HCVU651', EMPLOY='EMPLOY1')

sub_dat18 <- dat18 |> select(c("_STATE", "SEX1","PREGNANT", "MENTHLTH",
  ↪ "EMPLOY1", "_INCOMG", "EDUCA", "_HCVU651", "_RACE", "RENTHOM1",
  ↪ "MARITAL")) |> filter(SEX1==2) |> select(-SEX1) |>
  ↪ rename(INCOME='_INCOMG', INSURANCE='_HCVU651', EMPLOY='EMPLOY1')

```

merge relevant questionnaire years

```

data_list <- list(
  `2017` = sub_dat17,
  `2019` = sub_dat19,
  `2022` = sub_dat22,
  `2024` = sub_dat24,
  `2025` = sub_dat25
)

vars_to_keep <- c(
  "_STATE",
  "PREGNANT",
  "_AGEG5YR",
  "MENTHLTH",
  "EMPLOY",
  "INCOME",
  "EDUCA",
  "INSURANCE",
  "RACE",
  "MARITAL",
  "HOME",
  "FAMILY_PLANNING",

```

```

    "IYEAR",
    "IMONTH",
    "IDAY"
  )

analysis_data <- purrr::imap_dfr(
  data_list,
  function(dataset, year) {

    year <- as.integer(year)

    dataset <- dataset |>
      dplyr::select(dplyr::any_of(vars_to_keep)) |>
      dplyr::mutate(
        YEAR = year
      )

    if (year == 2022) {
      dataset <- dataset |>
        dplyr::mutate(
          IYEAR = as.integer(as.character(IYEAR)),
          IMONTH = as.integer(as.character(IMONTH)),
          IDAY = as.integer(as.character(IDAY)),

          date = as.Date(
            sprintf("%04d-%02d-%02d", IYEAR, IMONTH, IDAY)
          ),

          post_ban = dplyr::case_when(
            date < as.Date("2022-06-24") ~ 0L,
            date >= as.Date("2022-06-24") ~ 1L,
            TRUE ~ NA_integer_
          )
        )
    } else {
      dataset <- dataset |>
        dplyr::mutate(
          date = as.Date(NA),

          post_ban = dplyr::case_when(
            YEAR <= 2021 ~ 0L,
            YEAR >= 2023 ~ 1L,
            TRUE ~ NA_integer_
          )
        )
    }
  }
)

```

```

    )
  }
}
dataset
}
) |>
  dplyr::rename(STATE = `_STATE`) |> select(-c(IYEAR, IMONTH, IDAY, date))

# check that post ban was correctly created
analysis_data |> filter(YEAR == 2025) |> janitor::tabyl(post_ban)

```

post_ban	n	percent
1	187696	1

```
analysis_data |> filter(YEAR == 2024) |> janitor::tabyl(post_ban)
```

post_ban	n	percent
1	240183	1

```
analysis_data |> filter(YEAR == 2022) |> janitor::tabyl(post_ban)
```

post_ban	n	percent	valid_percent
0	96964	4.110508e-01	0.411056
1	138926	5.889365e-01	0.588944
NA	3	1.271763e-05	NA

```
analysis_data |> filter(YEAR == 2019) |> janitor::tabyl(post_ban)
```

post_ban	n	percent
0	228419	1

```
analysis_data |> filter(YEAR == 2017) |> janitor::tabyl(post_ban)
```

post_ban	n	percent
0	251007	1

create U.S. census regions

```
library(tibble)

census_region_lookup <- tribble(
  ~STATE, ~REGION, ~DIVISION,

  # Northeast
  "Connecticut",      "Northeast", "New England",
  "Maine",            "Northeast", "New England",
  "Massachusetts",    "Northeast", "New England",
  "New Hampshire",    "Northeast", "New England",
  "Rhode Island",     "Northeast", "New England",
  "Vermont",          "Northeast", "New England",
  "New Jersey",       "Northeast", "Middle Atlantic",
  "New York",         "Northeast", "Middle Atlantic",
  "Pennsylvania",     "Northeast", "Middle Atlantic",

  # Midwest
  "Illinois",         "Midwest", "East North Central",
  "Indiana",          "Midwest", "East North Central",
  "Michigan",         "Midwest", "East North Central",
  "Ohio",             "Midwest", "East North Central",
  "Wisconsin",        "Midwest", "East North Central",
  "Iowa",             "Midwest", "West North Central",
  "Kansas",           "Midwest", "West North Central",
  "Minnesota",        "Midwest", "West North Central",
  "Missouri",         "Midwest", "West North Central",
  "Nebraska",         "Midwest", "West North Central",
  "North Dakota",     "Midwest", "West North Central",
  "South Dakota",     "Midwest", "West North Central",

  # South
  "Delaware",         "South", "South Atlantic",
  "District of Columbia", "South", "South Atlantic",
  "Florida",          "South", "South Atlantic",
  "Georgia",          "South", "South Atlantic",
  "Maryland",         "South", "South Atlantic",
  "North Carolina",   "South", "South Atlantic",
  "South Carolina",   "South", "South Atlantic",
  "Virginia",         "South", "South Atlantic",
  "West Virginia",    "South", "South Atlantic",
```

```

"Alabama",           "South", "East South Central",
"Kentucky",          "South", "East South Central",
"Mississippi",       "South", "East South Central",
"Tennessee",         "South", "East South Central",
"Arkansas",          "South", "West South Central",
"Louisiana",         "South", "West South Central",
"Oklahoma",          "South", "West South Central",
"Texas",             "South", "West South Central",

# West
"Arizona",           "West", "Mountain",
"Colorado",          "West", "Mountain",
"Idaho",             "West", "Mountain",
"Montana",           "West", "Mountain",
"Nevada",            "West", "Mountain",
"New Mexico",        "West", "Mountain",
"Utah",              "West", "Mountain",
"Wyoming",           "West", "Mountain",
"Alaska",            "West", "Pacific",
"California",        "West", "Pacific",
"Hawaii",            "West", "Pacific",
"Oregon",            "West", "Pacific",
"Washington",        "West", "Pacific",

# Territories: not assigned to a Census region/division
"Guam",              NA_character_, NA_character_,
"Puerto Rico",       NA_character_, NA_character_,
"Virgin Islands",    NA_character_, NA_character_
)

# remove Guam, Puerto Rico, and Virgin Islands from analytic sample
analysis_data_v2 <- analysis_data |>
  dplyr::left_join(
    census_region_lookup,
    by = "STATE"
  ) |> filter(!is.na(REGION)) |> rename(AGEG5YR = "_AGEG5YR")

variable_labels <- list(
  AGEN5YR = c(
    `1` = "18 to 24 years",
    `2` = "25 to 29 years",

```

```

`3` = "30 to 34 years",
`4` = "35 to 39 years",
`5` = "40 to 44 years",
`6` = "45 to 49 years",
`7` = "50 to 54 years",
`8` = "55 to 59 years",
`9` = "60 to 64 years",
`10` = "65 to 69 years",
`11` = "70 to 74 years",
`12` = "75 to 79 years",
`13` = "80 years or older",
`14` = "Don't know/Refused/Missing"
),

MARITAL = c(
  `1` = "Married",
  `2` = "Divorced",
  `3` = "Widowed",
  `4` = "Separated",
  `5` = "Never married",
  `6` = "Member of an unmarried couple",
  `9` = "Refused"
),

HOME = c(
  `1` = "Own",
  `2` = "Rent",
  `3` = "Other arrangement",
  `7` = "Don't know",
  `9` = "Refused"
),

PREGNANT = c(
  `1` = "Yes",
  `2` = "No",
  `7` = "Don't know",
  `9` = "Refused"
),

MENTHLTH = c(
  setNames(
    paste(1:30, ifelse(1:30 == 1, "day", "days")),
    as.character(1:30)

```

```

    ),
    `77` = "Don't know",
    `88` = "None",
    `99` = "Refused"
  ),

  EMPLOY = c(
    `1` = "Employed for wages",
    `2` = "Self-employed",
    `3` = "Out of work for 1 year or more",
    `4` = "Out of work for less than 1 year",
    `5` = "Homemaker",
    `6` = "Student",
    `7` = "Retired",
    `8` = "Unable to work",
    `9` = "Refused"
  ),

  INCOME = c(
    `1` = "Less than $15,000",
    `2` = "$15,000 to $24,999",
    `3` = "$25,000 to $34,999",
    `4` = "$35,000 to $49,999",
    `5` = "$50,000 or more", # 2017, 2019 data only has $50k or more so align
    ↪ other years to this format
    `6` = "$50,000 or more",
    `7` = "$50,000 or more",
    `9` = "Don't know"
  ),

  EDUCA = c(
    `1` = "No high school",
    `2` = "High school",
    `3` = "Some college",
    `4` = "College graduate",
    `9` = "Don't know"
  ),

  INSURANCE = c(
    `1` = "Yes",
    `2` = "No",
    `9` = "Don't know"
  ),

```

```

RACE = c(
  `1` = "Non-Hispanic White",
  `2` = "Non-Hispanic Black",
  `3` = "Non-Hispanic Native American",
  `4` = "Non-Hispanic Asian",
  `5` = "Non-Hispanic Native Hawaiian",
  `6` = "Other race, non-Hispanic",
  `7` = "Non-Hispanic Multiracial",
  `8` = "Hispanic",
  `9` = "Don't know"
)
)

variables_to_summarize <- c(
  "MARITAL",
  "HOME",
  "PREGNANT",
  "MENTHLTH",
  "EMPLOY",
  "INCOME",
  "EDUCA",
  "INSURANCE",
  "RACE"
)

```

Assess Missingness by Age Group

6 = Age 45 to 49 (menopause starts around this age range) Pregnancy Status 1 = “Yes”, 2 = “No”, 7 = “Don’t know”, 9 = “Refused”

INSURANCE = Respondents aged 18-64 who have any form of health insurance (1=yes, 2=no, 9=don’t know)

Women who are 50 years old or older (code 7 and above) did not answer the pregnancy question.

```

analysis_data_v2 |>
  count(AGEG5YR, PREGNANT) |>
  group_by(AGEG5YR) |>
  mutate(percent = round(100 *n /sum(n),3)) |>
  ungroup()

```

```
# A tibble: 45 x 4
  AGE5YR PREGNANT      n percent
  <dbl>   <dbl> <int>   <dbl>
1     1     1     1  2254    4.01
2     1     2 51880   92.3
3     1     7    162    0.288
4     1     9     81    0.144
5     1    NA   1813    3.23
6     2     1   2872    5.77
7     2     2 45056   90.5
8     2     7    184    0.37
9     2     9    115    0.231
10    2    NA   1558    3.13
# i 35 more rows
```

```
analysis_data_v2 |>
  count(AGE5YR, INSURANCE) |>
  group_by(AGE5YR) |>
  mutate(percent = round(100 * n / sum(n), 3)) |>
  ungroup()
```

```
# A tibble: 32 x 4
  AGE5YR INSURANCE      n percent
  <dbl>   <dbl> <int>   <dbl>
1     1     1  45387   80.8
2     1     2   6234   11.1
3     1     9   4569    8.13
4     2     1 42106   84.6
5     2     2   6504   13.1
6     2     9   1175    2.36
7     3     1 50889   86.5
8     3     2   6917   11.8
9     3     9   1035    1.76
10    4     1 57814   88.1
# i 22 more rows
```

U.S. Population Characteristics Over Time

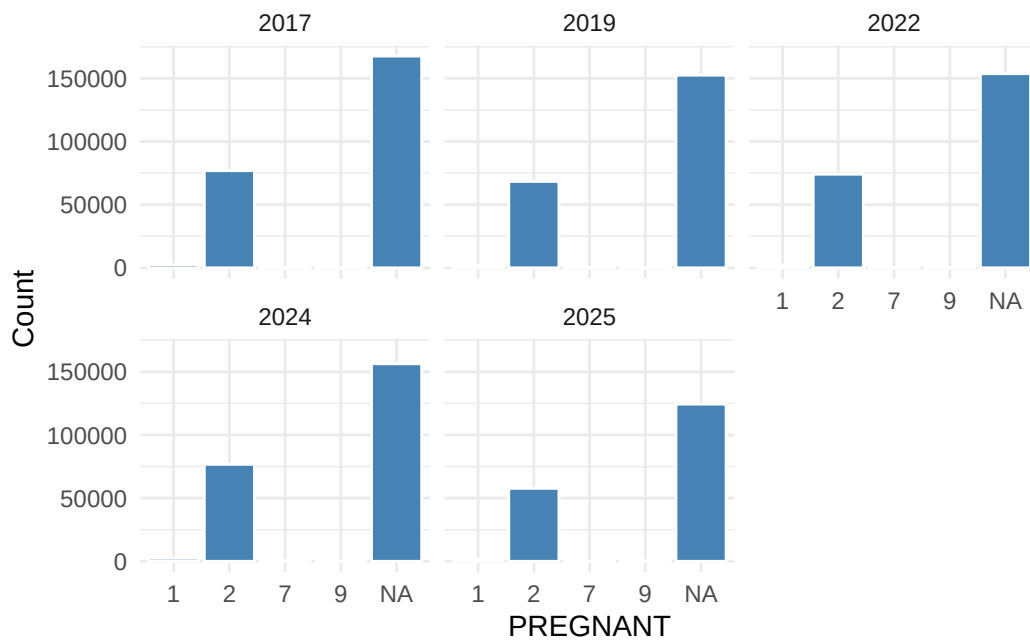
```
# categorical variables
sapply(names(analysis_data_v2)[!names(analysis_data_v2) %in%
  ↪ c("STATE", "YEAR", "post_ban", "DIVISION", "MENTHLTH", "REGION")],
  ↪ function(var){
```

```

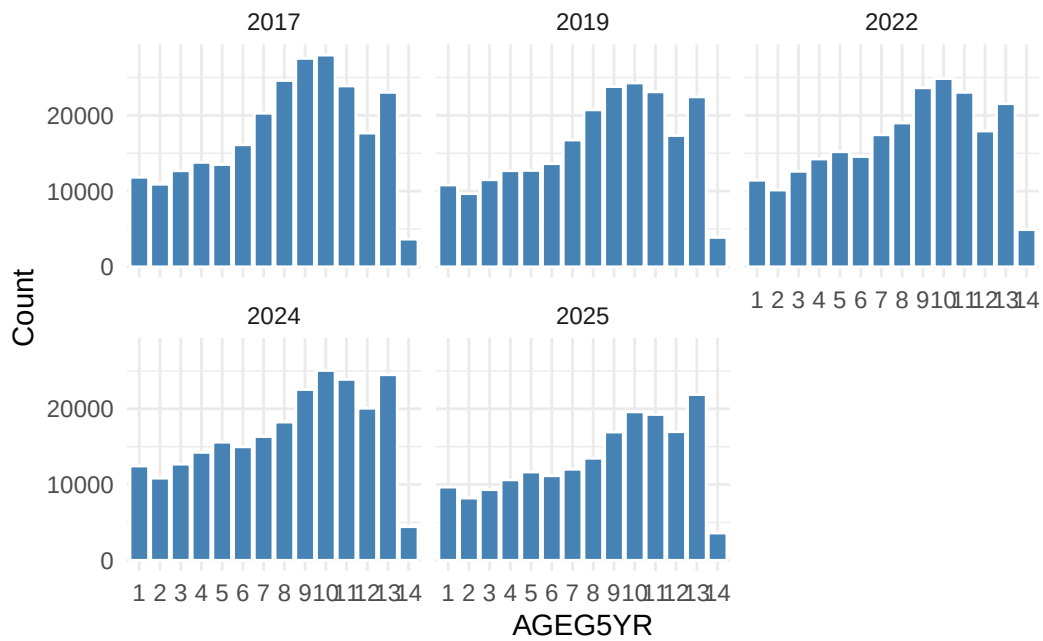
ggplot(
  analysis_data_v2,
  aes(x = factor(.data[[var]]))
) +
  geom_bar(
    fill = "steelblue",
    color = "white"
  ) +
  facet_wrap(~ YEAR) +
  theme_minimal() +
  labs(
    x = var,
    y = "Count"
  )
})

```

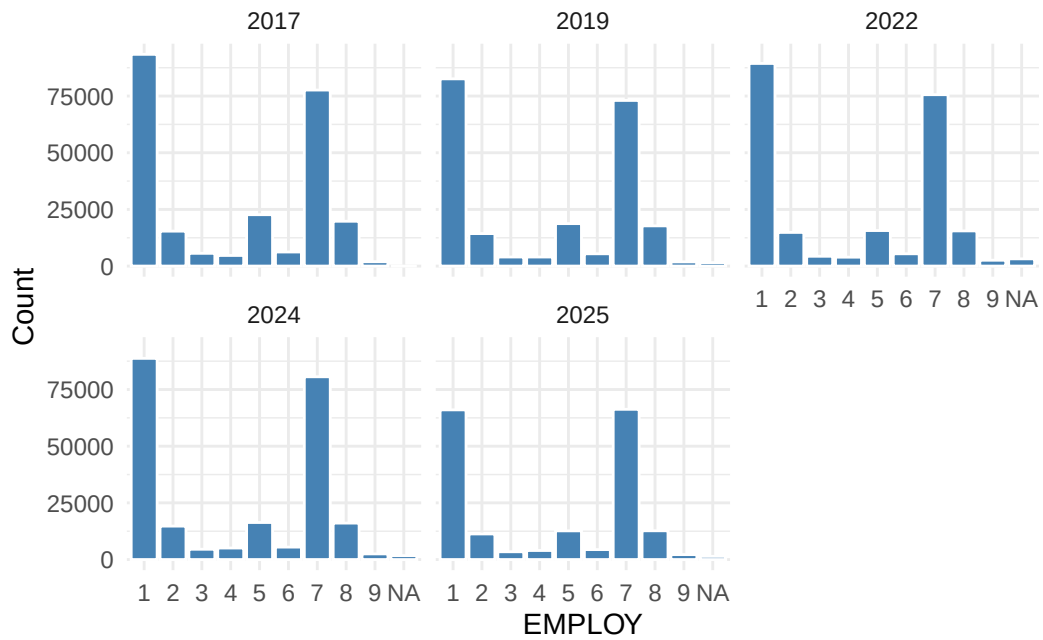
\$PREGNANT



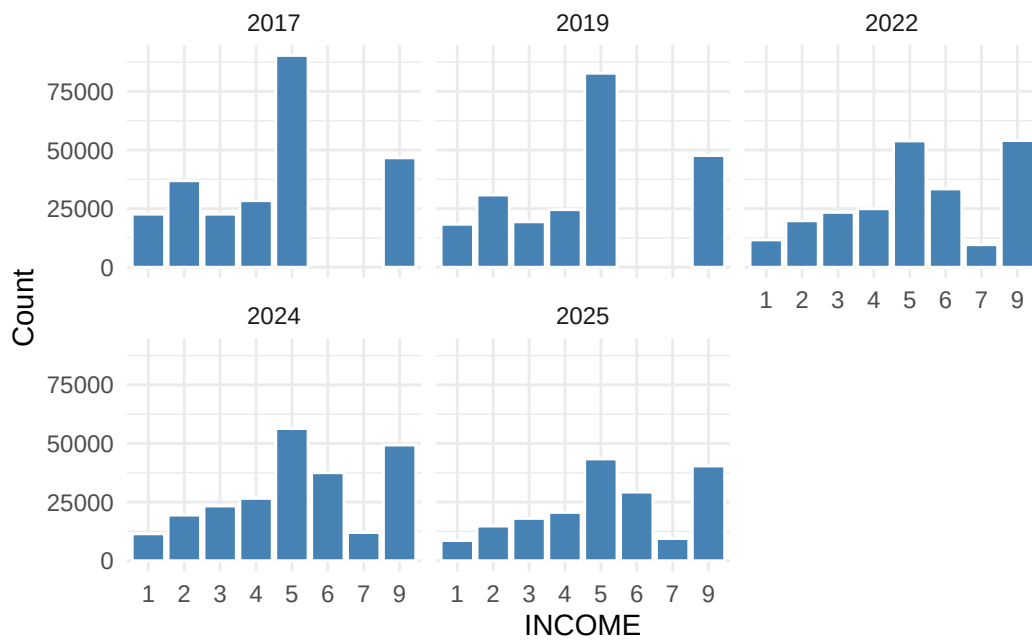
\$AGE5YR



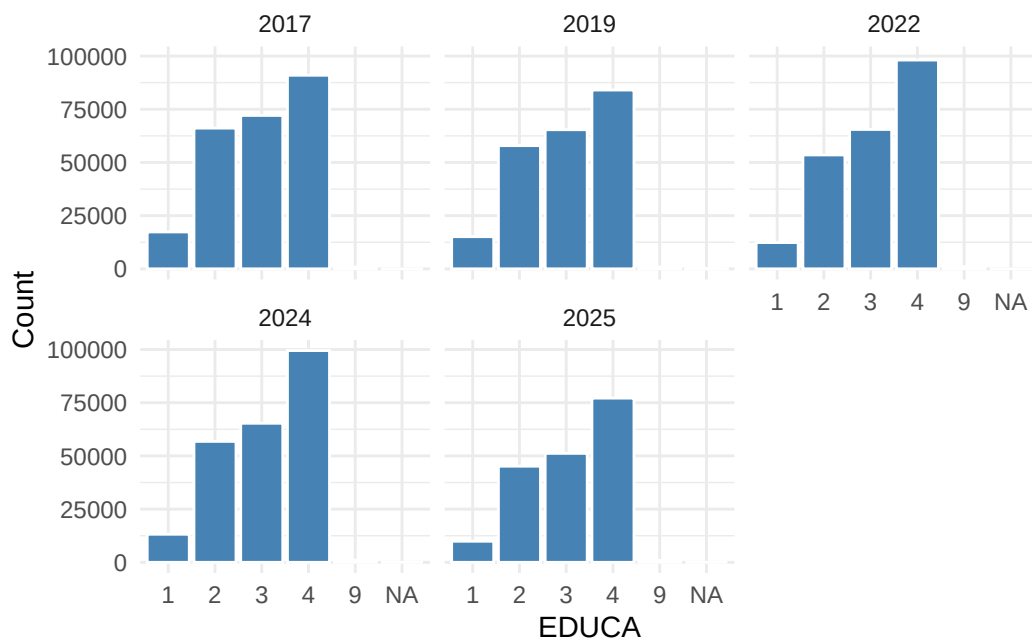
\$EMPLOY



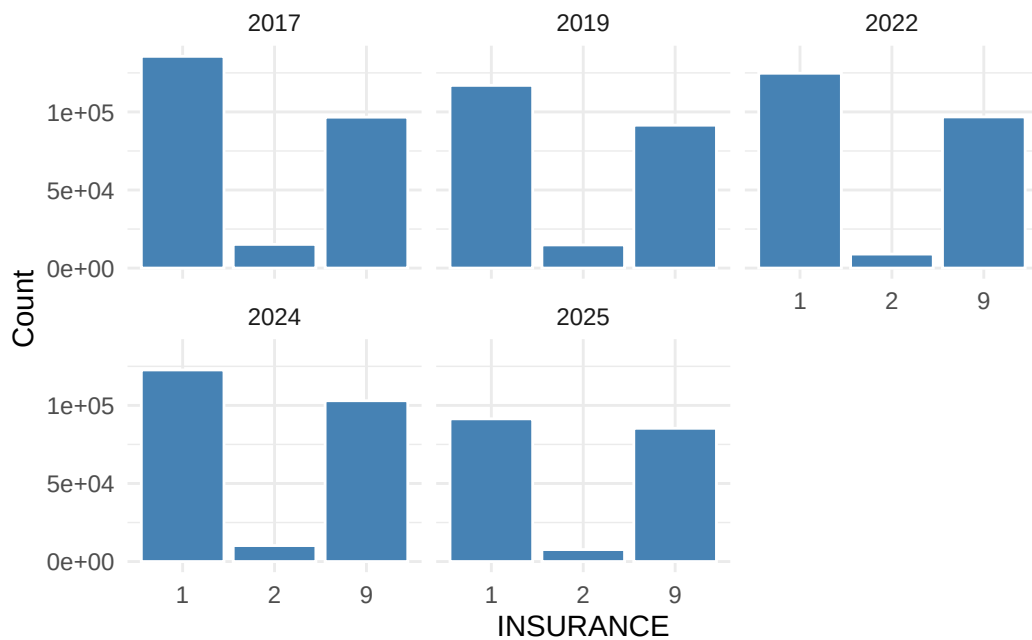
\$INCOME



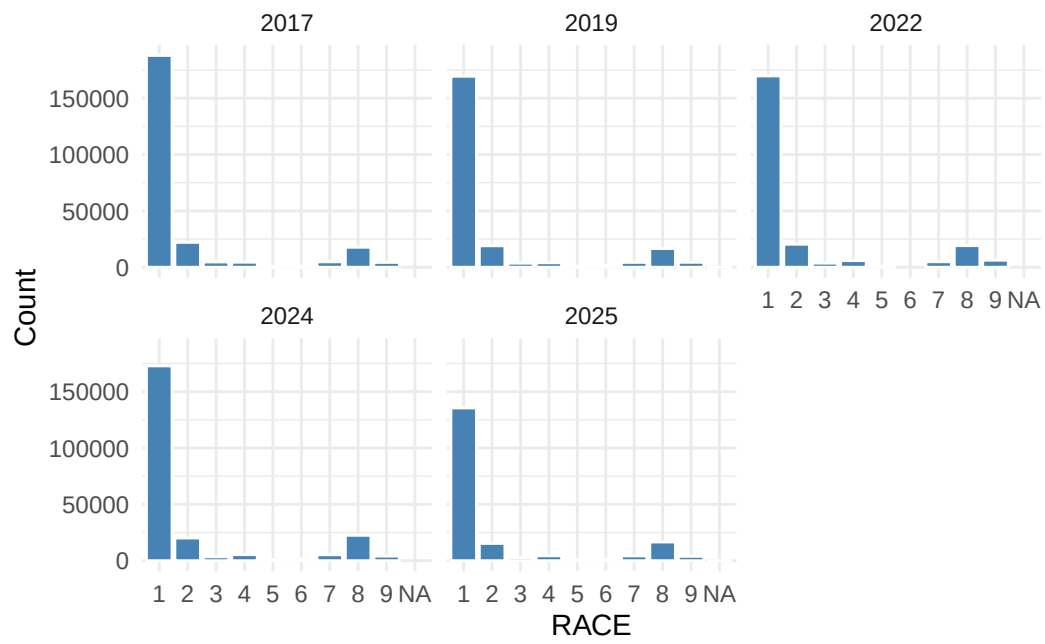
\$EDUCA



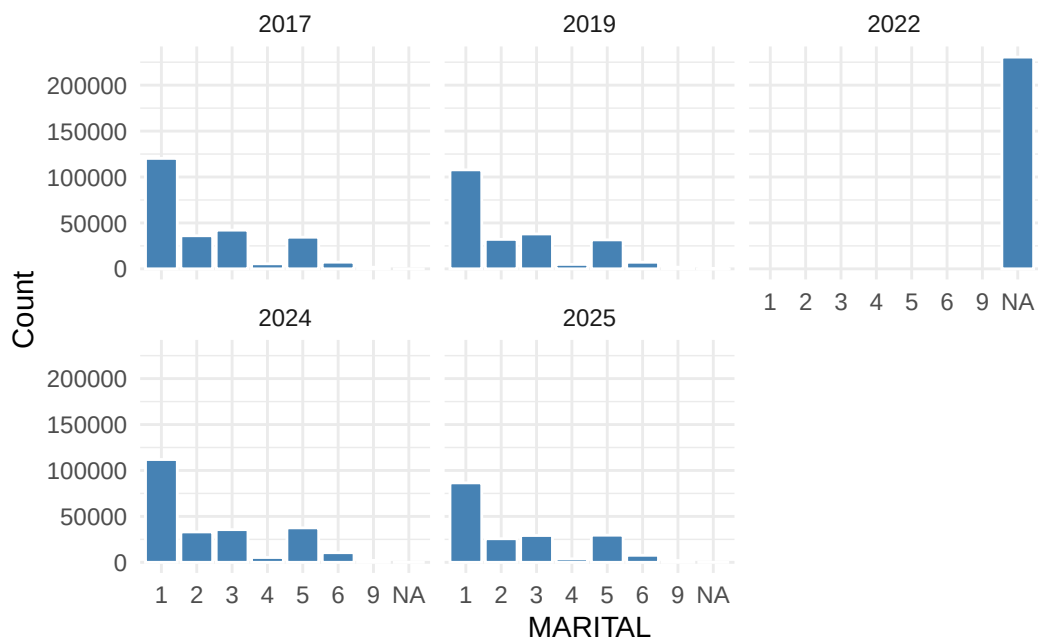
\$INSURANCE



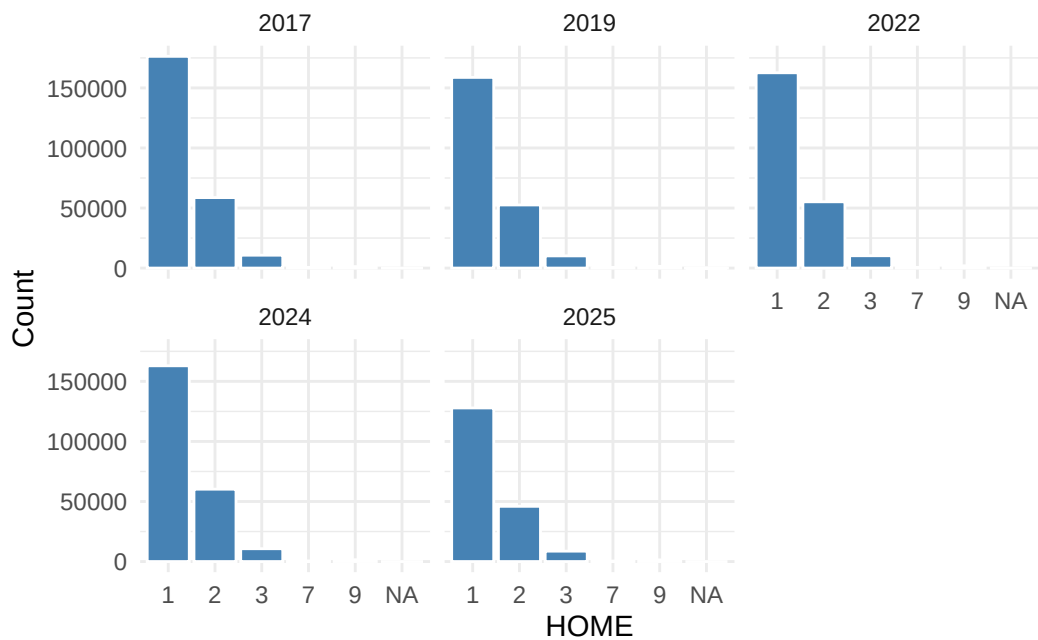
\$RACE



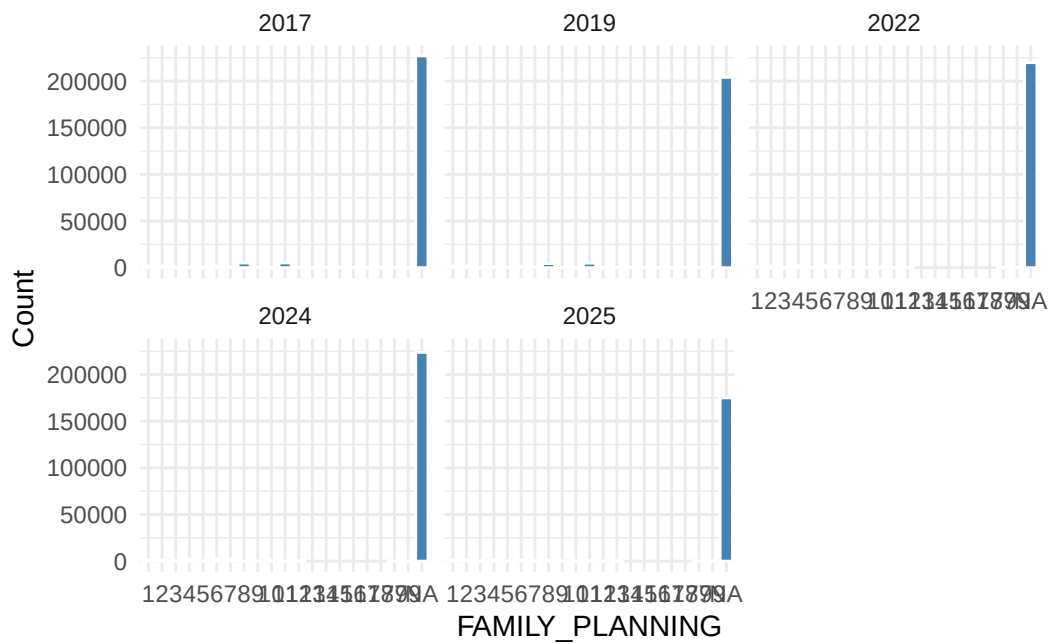
\$MARITAL



\$HOME



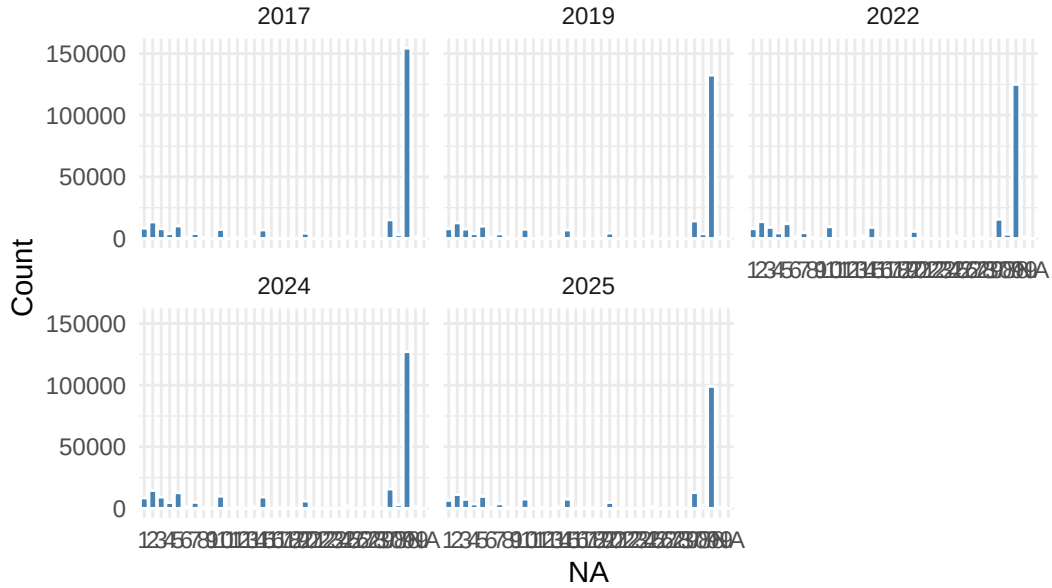
\$FAMILY_PLANNING



```
# numerical variable
ggplot(
  analysis_data_v2 ,
  aes(x = factor(MENTHLTH))
) +
  geom_bar(
    fill = "steelblue",
    color = "white"
  ) +
  facet_wrap(~ YEAR) +
  theme_minimal() +
  labs(
    title = "Distribution of Days with Poor Mental Health",
    x = var,
    y = "Count"
  )
```

Warning in label(labels[[nm]] %||% ""): NAs introduced by coercion

Distribution of Days with Poor Mental Health



```
get_descriptive_stats_over_entire_period <- function(data, var) {

  tab <- data |>
    janitor::tabyl(
      {{ var }},
      show_na = TRUE
    )

  # The first column is the categorical variable
  category_col <- names(tab)[1]

  total_n <- sum(tab$n)

  valid_n <- sum(
    tab$n[!is.na(tab[[category_col]])]
  )

  tab |>
    dplyr::mutate(
      percent = 100 * n / total_n,
      valid_percent = dplyr::if_else(
        valid_n > 0 & !is.na(.data[[category_col]]),
        100 * n / valid_n,
```

```

        NA_real_
    )
}
get_descriptive_stats_over_entire_period(analysis_data,PREGNANT)

```

PREGNANT	n	percent	valid_percent
1	11432	1.00000175	3.0394636
2	362558	31.71436619	96.3944922
7	947	0.08283779	0.2517820
9	1182	0.10339416	0.3142622
NA	767079	67.09940010	NA

```

get_descriptive_stats_over_entire_period(analysis_data,MENTHLTH)

```

MENTHLTH	n	percent	valid_percent
1	39555	3.460030546	3.46006384
2	66084	5.780625928	5.78068155
3	41539	3.633578785	3.63361375
4	20903	1.828467160	1.82848475
5	55130	4.822436708	4.82248311
6	5731	0.501312983	0.50131781
7	21584	1.888036893	1.88805506
8	4351	0.380598986	0.38060265
9	731	0.063943429	0.06394404
10	42260	3.696647475	3.69668304
11	300	0.026242173	0.02624243
12	3036	0.265570794	0.26557335
13	421	0.036826516	0.03682687
14	8309	0.726820726	0.72682772
15	39977	3.496944536	3.49697818
16	697	0.060969316	0.06096990
17	601	0.052571820	0.05257233
18	785	0.068667020	0.06866768
19	128	0.011196661	0.01119677
20	25415	2.223149446	2.22317084
21	1411	0.123425688	0.12342688
22	400	0.034989564	0.03498990
23	271	0.023705430	0.02370566
24	259	0.022655743	0.02265596

25	8695	0.760585655	0.76059297
26	269	0.023530482	0.02353071
27	593	0.051872029	0.05187253
28	2362	0.206613378	0.20661537
29	1140	0.099720258	0.09972122
30	75139	6.572702192	6.57276544
77	15937	1.394071718	1.39408513
88	653219	57.139620608	57.14017042
99	5955	0.520907139	0.52091215
NA	11	0.000962213	NA

```
get_descriptive_stats_over_entire_period(analysis_data,EMPLOY)
```

EMPLOY	n	percent	valid_percent
1	427657	37.4088303	37.6710651
2	72200	6.3156164	6.3598887
3	22839	1.9978167	2.0118213
4	22202	1.9420958	1.9557098
5	90549	7.9206752	7.9761989
6	27679	2.4211904	2.4381629
7	377824	33.0497429	33.2814207
8	83087	7.2679448	7.3188929
9	11203	0.9799702	0.9868398
NA	7958	0.6961174	NA

```
get_descriptive_stats_over_entire_period(analysis_data,INCOME)
```

INCOME	n	percent	valid_percent
1	77835	6.808532	6.808532
2	125785	11.002906	11.002906
3	108922	9.527833	9.527833
4	126922	11.102364	11.102364
5	329376	28.811807	28.811807
6	100829	8.819907	8.819907
7	31201	2.729273	2.729273
9	242328	21.197378	21.197378

```
get_descriptive_stats_over_entire_period(analysis_data,EDUCA)
```

EDUCA	n	percent	valid_percent
1	70463	6.163674184	6.1638144
2	285253	24.952195508	24.9527630
3	324998	28.428846097	28.4294927
4	457453	40.015202966	40.0161131
9	5005	0.437806924	0.4378169
NA	26	0.002274322	NA

```
get_descriptive_stats_over_entire_period(analysis_data, INSURANCE)
```

INSURANCE	n	percent	valid_percent
1	605179	52.937374	52.937374
2	58134	5.085208	5.085208
9	479885	41.977418	41.977418

Summarize Categorical Variables into Count and Percentages

```
library(dplyr)
library(tidyr)
library(purrr)

summarize_categorical <- function(
  data,
  group_vars,
  response_vars,
  variable_labels
) {

  group_vars <- as.character(group_vars)
  response_vars <- as.character(response_vars)

  required_vars <- c(group_vars, response_vars)
  missing_vars <- setdiff(required_vars, names(data))

  if (length(missing_vars) > 0) {
    stop(
      "These variables are not in the dataset: ",
      paste(missing_vars, collapse = ", ")
    )
  }
}
```

```

# Function to retrieve labels
lookup_label <- function(variable, code) {

  # Actual NA values
  if (is.na(code) || code == "") {
    return("Missing")
  }

  labels <- variable_labels[[variable]]

  # No label vector exists for the variable
  if (is.null(labels)) {
    return(paste("Unlabeled variable:", variable))
  }

  label <- unname(labels[as.character(code)])

  # Code does not exist in the label vector
  if (length(label) == 0 || is.na(label)) {
    return(paste("Unlabeled code:", code))
  }

  label
}

output_data <- data |>
  select(all_of(required_vars))

# Create labeled versions of subgroup variables
subgroup_label_vars <- character(0)

for (group_var in group_vars) {

  if (group_var %in% names(variable_labels)) {

    label_var <- paste0(group_var, "_LABEL")

    output_data[[label_var]] <- purrr::map_chr(
      as.character(output_data[[group_var]]),
      ~ lookup_label(group_var, .x)
    )
  }
}

```

```

    subgroup_label_vars <- c(subgroup_label_vars, label_var)
  }
}

subgroup_vars_for_grouping <- c(
  group_vars,
  subgroup_label_vars
)

output_data |>
  pivot_longer(
    cols = all_of(response_vars),
    names_to = "VARIABLE",
    values_to = "VALUE"
  ) |>
  mutate(
    VALUE_CODE = if_else(
      is.na(VALUE),
      "Missing",
      as.character(VALUE)
    ),

    VALUE_LABEL = purrr::map2_chr(
      VARIABLE,
      VALUE_CODE,
      lookup_label
    )
  ) |>
  group_by(
    across(
      all_of(
        c(
          subgroup_vars_for_grouping,
          "VARIABLE",
          "VALUE_CODE",
          "VALUE_LABEL"
        )
      )
    )
  ) |>
  summarise(
    N = n(),
    .groups = "drop"
  )

```

```

) |>
group_by(
  across(
    all_of(
      c(
        subgroup_vars_for_grouping,
        "VARIABLE"
      )
    )
  )
) |>
mutate(
  PERCENT = round(100 * N / sum(N), 2)
) |>
ungroup() |>
arrange(
  across(all_of(subgroup_vars_for_grouping)),
  VARIABLE,
  VALUE_CODE
)
}

```

show summaries for any variable

```

summarize_categorical(
  data = analysis_data_v2 |> filter(AGEG5YR < 7), # filter out women 50 year
  ↪ and above to reduce missingness in pregnant and insurance response
  group_vars = c(
    "STATE",
    "REGION",
    "YEAR"
  ),
  response_vars = "INSURANCE", # change this variable to any you want
  variable_labels = variable_labels
)

```

A tibble: 749 x 8

	STATE	REGION	YEAR	VARIABLE	VALUE_CODE	VALUE_LABEL	N	PERCENT
	<chr>	<chr>	<int>	<chr>	<chr>	<chr>	<int>	<dbl>
1	Alabama	South	2017	INSURANCE	1	Yes	1053	83.4

2	Alabama	South	2017	INSURANCE	2	No	208	16.5
3	Alabama	South	2017	INSURANCE	9	Don't know	2	0.16
4	Alabama	South	2019	INSURANCE	1	Yes	1071	82.8
5	Alabama	South	2019	INSURANCE	2	No	221	17.1
6	Alabama	South	2019	INSURANCE	9	Don't know	2	0.15
7	Alabama	South	2022	INSURANCE	1	Yes	682	87.7
8	Alabama	South	2022	INSURANCE	2	No	79	10.2
9	Alabama	South	2022	INSURANCE	9	Don't know	17	2.19
10	Alabama	South	2024	INSURANCE	1	Yes	898	90.3

i 739 more rows

```
plot_variable <- function(data, variable_name, plot_title, legend_title) {
  data |>
    dplyr::filter(VARIABLE == variable_name) |>
    ggplot(
      aes(
        x = factor(YEAR),
        y = PERCENT,
        fill = VALUE_LABEL
      )
    ) +
    geom_col(position = "dodge") +
    facet_wrap(~ REGION) +
    labs(
      title = plot_title,
      x = "Year",
      y = "Percent",
      fill = legend_title
    ) +
    theme_minimal() +
    theme(
      legend.position = "bottom",
      axis.text.x = element_text(angle = 45, hjust = 1)
    )
}
```

Trends in the U.S. Regions Over Time

```
var_by_region_plots <- list(
```

```

marital = plot_variable(
  summarize_categorical(
    data = analysis_data_v2 |> filter(AGEG5YR < 7),
    group_vars = c(
      "STATE",
      "REGION",
      "YEAR"
    ),
    response_vars = "MARITAL",
    variable_labels = variable_labels
  ),
  "MARITAL",
  "Marital Status by Region and Year",
  "Marital status"
),

home = plot_variable(
  data = summarize_categorical(
    data = analysis_data_v2 |> filter(AGEG5YR < 7),
    group_vars = c(
      "STATE",
      "REGION",
      "YEAR"
    ),
    response_vars = "HOME",
    variable_labels = variable_labels
  ),
  "HOME",
  "Homeowner Status by Region and Year",
  "Homeowner status"
),

pregnant = plot_variable(
  data = summarize_categorical(
    data = analysis_data_v2 |> filter(AGEG5YR < 7),
    group_vars = c(
      "STATE",
      "REGION",
      "YEAR"
    ),
    response_vars = "PREGNANT",
    variable_labels = variable_labels
  ),
  "PREGNANT",
  "Pregnant Status by Region and Year",
  "Pregnant status"
)

```

```

),
  "PREGNANT",
  "Pregnancy Status by Region and Year",
  "Pregnancy status"
),

employ = plot_variable(
  data = summarize_categorical(
data = analysis_data_v2 |> filter(AGEG5YR < 7),
group_vars = c(
  "STATE",
  "REGION",
  "YEAR"
),
  response_vars = "EMPLOY",
  variable_labels = variable_labels
),
  "EMPLOY",
  "Employment Status by Region and Year",
  "Employment status"
),

income = plot_variable(
  data = summarize_categorical(
data = analysis_data_v2 |> filter(AGEG5YR < 7),
group_vars = c(
  "STATE",
  "REGION",
  "YEAR"
),
  response_vars = "INCOME",
  variable_labels = variable_labels
),
  "INCOME",
  "Income Categories by Region and Year",
  "Income category"
),

insurance = plot_variable(
  data = summarize_categorical(
data = analysis_data_v2 |> filter(AGEG5YR < 7),
group_vars = c(
  "STATE",

```

```

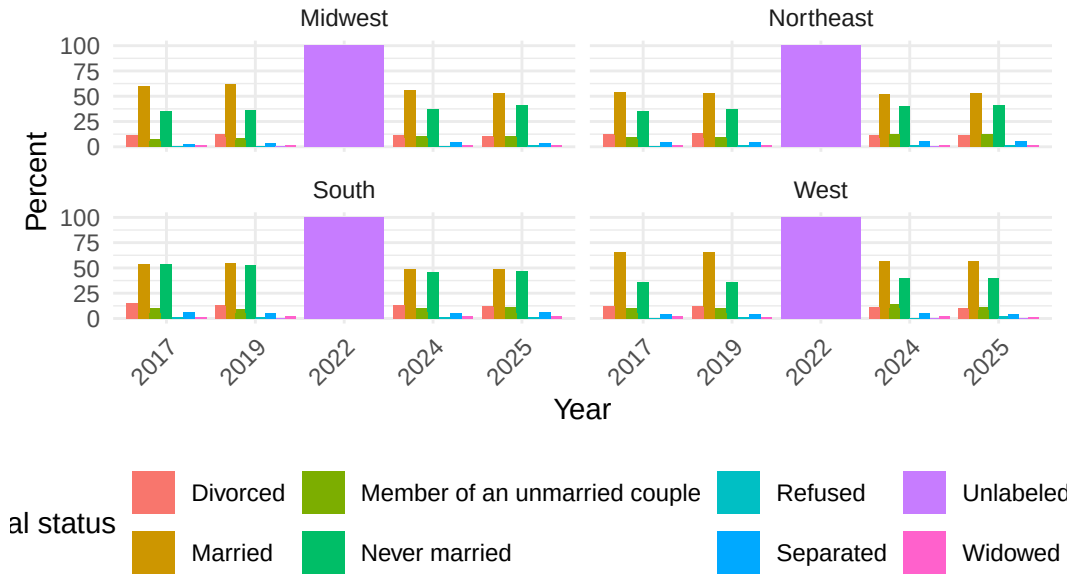
    "REGION",
    "YEAR"
  ),
  response_vars = "INSURANCE",
  variable_labels = variable_labels
),
  "INSURANCE",
  "Health Insurance Status by Region and Year",
  "Insurance status"
),

race = plot_variable(
  data = summarize_categorical(
data = analysis_data_v2 |> filter(AGEG5YR < 7),
group_vars = c(
  "STATE",
  "REGION",
  "YEAR"
),
  response_vars = "RACE",
  variable_labels = variable_labels
),
  "RACE",
  "Race and Ethnicity by Region and Year",
  "Race and ethnicity"
)
)

```

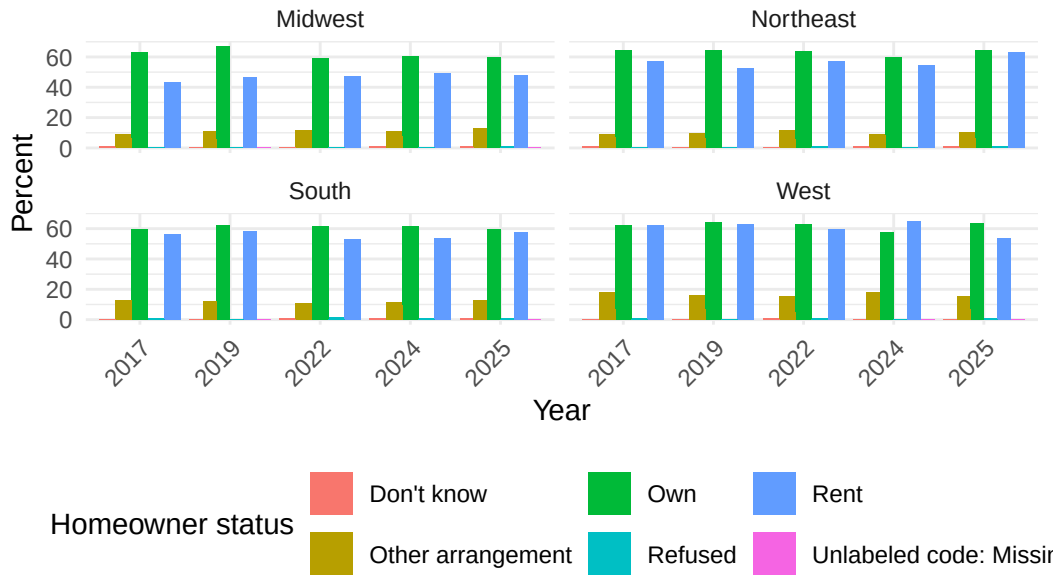
```
var_by_region_plots$marital
```

Marital Status by Region and Year

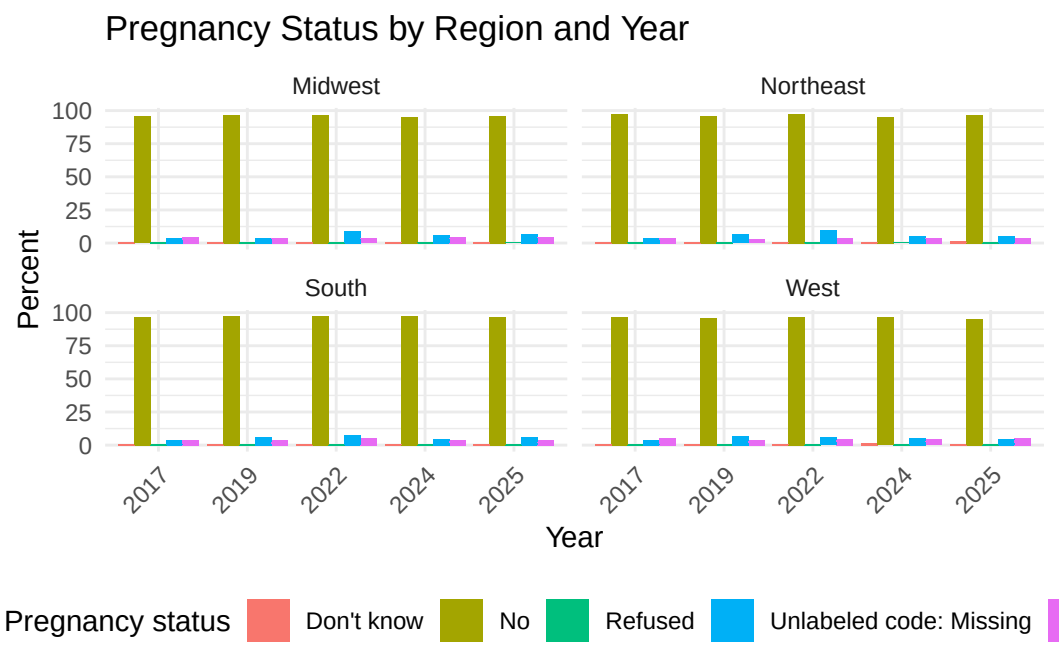


var_by_region_plots\$home

Homeowner Status by Region and Year



```
var_by_region_plots$pregnant
```



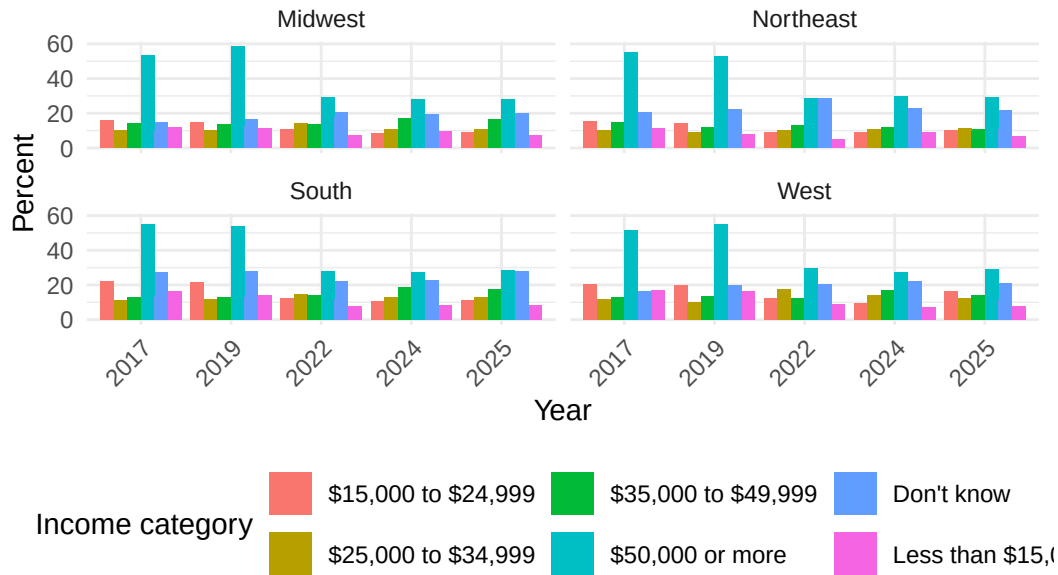
```
var_by_region_plots$employ
```

Employment Status by Region and Year

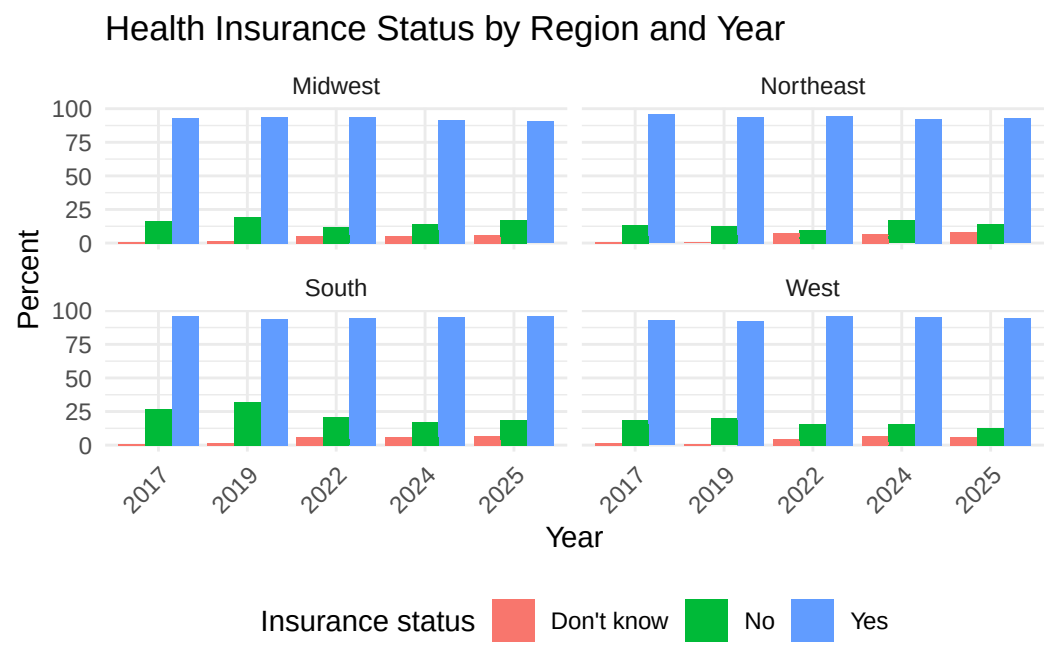


```
var_by_region_plots$income
```

Income Categories by Region and Year

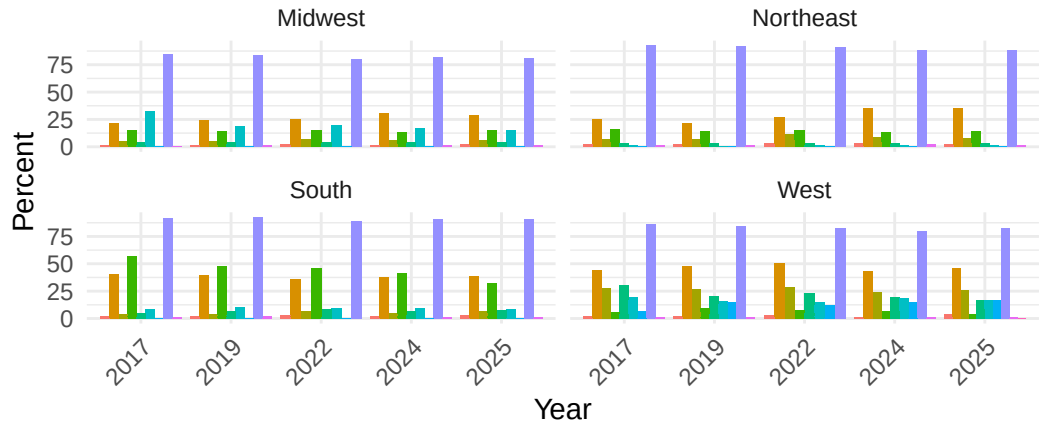


```
var_by_region_plots$insurance
```



```
var_by_region_plots$race
```

Race and Ethnicity by Region and Year



know Non-Hispanic Asian Non-Hispanic Multiracial Non-Hispanic Native
 nic Non-Hispanic Black Non-Hispanic Native American Non-Hispanic White